

How to give read and write permissions using Unix commands.
A very simple example is to create a text file with a C program,
then use chmod.

Create a file called permission.c

```
#include <stdio.h>
int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");
    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }
    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);
    printf("File created successfully.\n");
    return 0;
}
```

Compile

```
gcc permission.c -o permission
```

Run

```
./permission
```

Output

File created successfully.

A file called sample.txt is created.

Check the current permissions

```
ls -l sample.txt
```

You may see

```
-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt
```

The owner has: Read + Write

Group has: Read

Others have: Read

This is permission 644.

Give Read and Write permission to everyone

```
chmod 666 sample.txt
```

Check `ls -l sample.txt`

You should see

```
-rw-rw-rw- 1 user user ... sample.txt
```

Instead of numbers, you can directly give:

```
chmod u+rw sample.txt
```

chmod o+rw sample.txt

chmod a+rw sample.txt

Task of the Day

- 1) Give Read and Write only to the owner
- 2) Give Read and Write to the owner and group
- 3) Only owner can read and write
- 4) Owner read/write, everyone else read-only
- 5) Owner can do everything, others can only read and execute
- 6) Owner and group can read/write
- 7) Only owner can execute

Fill in the permission numbers for the following:

- 8) Owner: read + write; Group: nothing; Others: nothing
→600
- 9) Owner: read + write; Group: read; Others: read →644

-
- 10) Owner: read + write + execute; Group: read + execute;
Others: read + execute → 755
 - 11) _____ Owner: read + write; Group: read
+ write; Others: nothing →660
 - 12) Everyone: read only →


```
boddanapurnachandarrao@boddanas-MacBook-Air Permission % chmod u+rw sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % ls -l sample.txt
-rwxrw-rw- 1 boddanapurnachandarrao staff 28 17 Aug 10:37 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % chmod o+rw sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % ls -l sample.txt
-rwxrw-rw- 1 boddanapurnachandarrao staff 28 17 Aug 10:37 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % chmod a+rw sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % ls -l sample.txt
-rwxrw-rw- 1 boddanapurnachandarrao staff 28 17 Aug 10:37 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % chmod 600 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % ls -l sample.txt
-rw----- 1 boddanapurnachandarrao staff 28 17 Aug 10:37 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % chmod 660 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % ls -l sample.txt
-rw-rw---- 1 boddanapurnachandarrao staff 28 17 Aug 10:37 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % chmod 600 sample.txt
zsh: command not found: chmod
boddanapurnachandarrao@boddanas-MacBook-Air Permission % chmod 600 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % ls -l sample.txt
-rw----- 1 boddanapurnachandarrao staff 28 17 Aug 10:37 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % chmod 644 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % ls -l sample.txt
-rw-r--r-- 1 boddanapurnachandarrao staff 28 17 Aug 10:37 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % chmod 755 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % ls -l sample.txt
-rwxr-xr-x 1 boddanapurnachandarrao staff 28 17 Aug 10:37 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % chmod 660 sample.txt
boddanapurnachandarrao@boddanas-MacBook-Air Permission % ls -l sample.txt
```